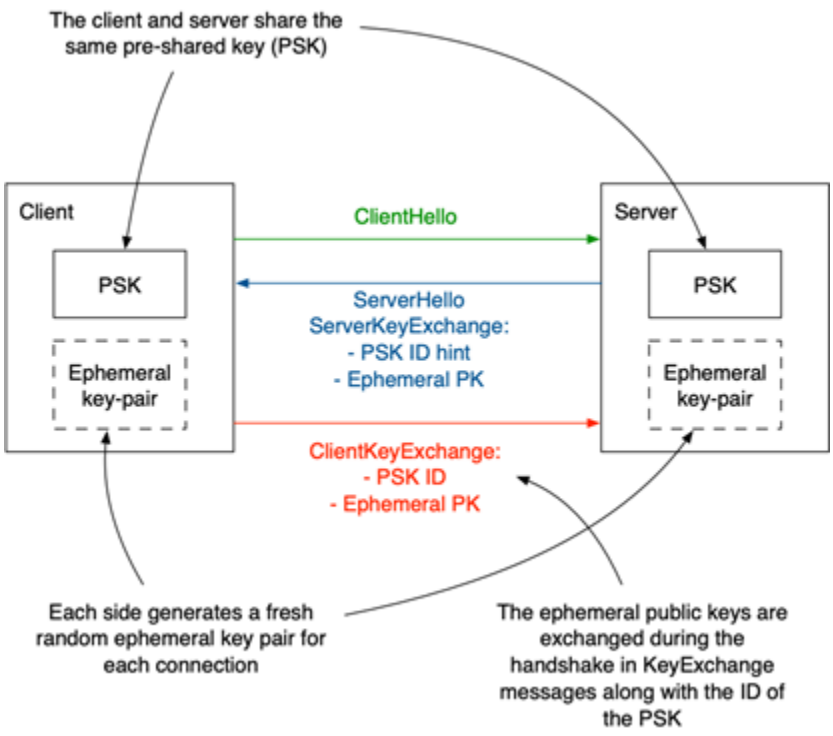


Figure 12.6 PSK cipher suites with forward secrecy use ephemeral key pairs in addition to the PSK. The client and server swap ephemeral public keys in key exchange messages during the TLS handshake. A Diffie-Hellman key agreement is then performed between each side's ephemeral private key and the received ephemeral public, which produces an identical secret value that is then mixed into the TLS key derivation process.

Custom protocols and the Noise protocol framework

Although for most IoT applications TLS or DTLS should be perfectly adequate for your needs, you may feel tempted to design your own cryptographic protocol that is a custom fit for your application. This is almost always a mistake, as even experienced cryptographers have made serious mistakes when designing protocols. Despite this widely repeated advice, many custom IoT security protocols have been developed, and new ones continue to be made. If you feel that you must develop a custom protocol for your application and can't use TLS or DTLS, the Noise protocol framework (https://noiseprotocol.org) can be used as a